STAD70 Course Work

Richard Ye

Contents

1 Introduction **5**

2 Financial Data and Returns **7**

2.1 Trading 8

2.2 Order Types 8

2.3 Returns 8

3 Return Distributions **11**

3.1 Skewness 11

3.2 Kurtosis 12

3.3 QQ Plot 13

3.4 Heavy Tail Distributions 13

3.5 Mixture Distributions 16

4 Problem Set 1 **17**

4.1 Q1 17

4.2 Q2 20

4.3 Q3 24

4.4 Q4 27

5 Problem Set 2 **29**

5.1 Q1 29

5.2 Q2 30

5.3 Q3 31

Chapter 1

Introduction

Work from the STAD70 course taken Winter 2023.

Chapter 2

Financial Data and Returns

- Prices of financial instruments (stocks, bonds, futures, options, etc)
- LIBOR: London Inter-Bank Offered Rate
  - Avg interest rate that major London Banks would charge when borrowing from each other
- FX Rates: Decentralized market that sets currency prices

There are two types of *raw (tick)* data:

- **Quote data**
  - Record of bid/ask prices from the order book, usually just the top N lines of the order book on the bid and ask sides
- **Trade data**
  - Trade records (filled orders)

Data that is identified using time:

- **Intraday Data**
  - Data of the most current bids/asks, last price, volume etc
  - Describes the data of a stock within a given day
- **Daily data**
  - For longer term analysis

2.1 Trading

There are two types: - **OTC (Over the counter)** - Negotiated and traded directly between parties, not for the public to bid on - **Through an Exchange** - The TSX, NYSE, LSE (London) - Uses an **auction** system where there are many potential buyers and sellers - Uses a **continuous double auction (CDA)**, where the bids and asks are matched in real-time to determine which trades to execute. - A matching order (bid $\geq$ ask) is executed right away - Outstanding orders are maintained in an **order book**

2.2 Order Types

- Limit order: buy/sell are no more/les than a specified price
  - It is maintained in the order book if not filled immediately
- Market order: buy/sell at the current market price immediately
  - No control over the price at which the order will execute
- Iceberg order: Only a portion of the total (giant) order is displayed in the order book, and when it is filled, a new portion is shown
  - Used to maintain anonymity

2.3 Returns

- The ratio of money gained/lost on an investment relative to the invested amount
- It is defined relative tot he holding period (daily, monthly, annual)

2.3.1 Net Returns

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} = \frac{P_t}{P_{t-1}} - 1$$

Where P_t is the price at time T.

Gross return is just $\frac{P_t}{P_{t-1}} = 1 + R_t$

2.3.2 Log Returns

$$\begin{aligned} r_t &= \log(1 + R_t) \\ &= \log\left(\frac{P_t}{P_{t-1}}\right) \\ &= \log(P_t) - \log(P_{t-1}) \end{aligned}$$

You can obtain the log return from the net return (for small returns, $< 1\%$) using the Taylor Approximation:

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2}f''(x_0)(x - x_0)^2 + ...$$

$$\begin{aligned} r_t &= \log(1 + R_t) \\ &\approx \log(x_0) + \log'(x)|_{x=x_0} \cdot (\underbrace{1 + R_t}_{x} - x_0) + ... \\ &\text{Expand around } x_0 = 1 \\ &= 0 + \frac{1}{1} \cdot R_t \\ &\approx R_t \end{aligned}$$

We like to work with log returns because they are also easy to aggregate!

$$\begin{aligned} R_{1-22} &= (1 + R_1) \cdot (1 + R_2) \cdot \dots \cdot (1 + R_{22}) - 1 \\ r_{1-22} &= r_1 + r_2 + \dots + r_{22} \\ &= \log\left(\frac{P_1}{P_0} \cdot \frac{P_2}{P_1} \cdot \dots \cdot \frac{P_{22}}{P_{21}}\right) \\ &= \log\left(\frac{P_{22}}{P_0}\right) \end{aligned}$$

2.3.3 Returns accounting for dividends

If a dividend was paid just before time t, and after $t - 1$, then we would have:

$$R_t = \frac{P_t - D_t}{P_{t-1}} - 1$$

$$r_t = \log(P_t + D_t) - \log(P_{t-1})$$

2.3.4 Returns accounting for splits

$$R_t = \frac{P_t}{P_{t-1}/s} - 1$$
$$r_t = \log(P_t) - \log(P_{t-1}/s)$$
$$\text{Where } s = \text{number of shares recieved per 1 share owned}$$

2.3.5 Adjusted returns

These returns have already accounted for dividends and splits, so returns should always be calculated on **adjusted returns. They should not be used as prices (ie the adjusted close)!**

2.3.6 Random Walk Model

We say the additive log returns

$$\log\left(\frac{P_t}{P_0}\right) = r_1 + r_2 + \cdots + r_t$$

Follow a random walk model if:

$$r_t \sim (\mu, \sigma^2)$$

$$\text{Where} \quad \mathbb{E}[r_i] = \mu \text{ is drift}$$
$$\sqrt{Var[r_i]} = \sigma \text{ is volatility}$$

However they follow a Normal random walk if:

$$r_t \sim N(\mu, \sigma^2)$$

Which implies that log-returns are Normally distributed.

The random walk from $1 - n$ has:

$$r_{1-n} = r_1 + r_2 + \cdots + r_n$$

$$\text{Mean: } \mathbb{E}(r_{1-n}) = n\mu$$
$$\text{Variance: } \sqrt{Var[r_{1-n}]} = \sqrt{n}\sigma$$

Which can be converted to the asset price simply using

$$P_t = P_0 \cdot e^{r_1 + r_2 + \cdots + r_t}$$

Which is referred to as the Exponential/Geometric Random walk

Random walks are not a good description of reality, but they are useful for modelling.

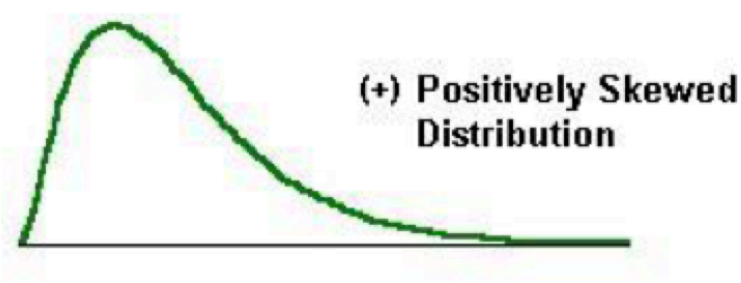
Chapter 3

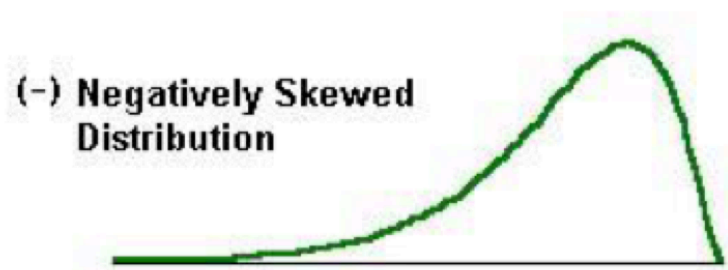
Return Distributions

3.1 Skewness

Skewness measures symmetry. - Positive skew → right skewed - Negative skew → left skewed

$$Sk = \mathbb{E}\left[\left(\frac{X-\mu}{\sigma}\right)^3\right]$$



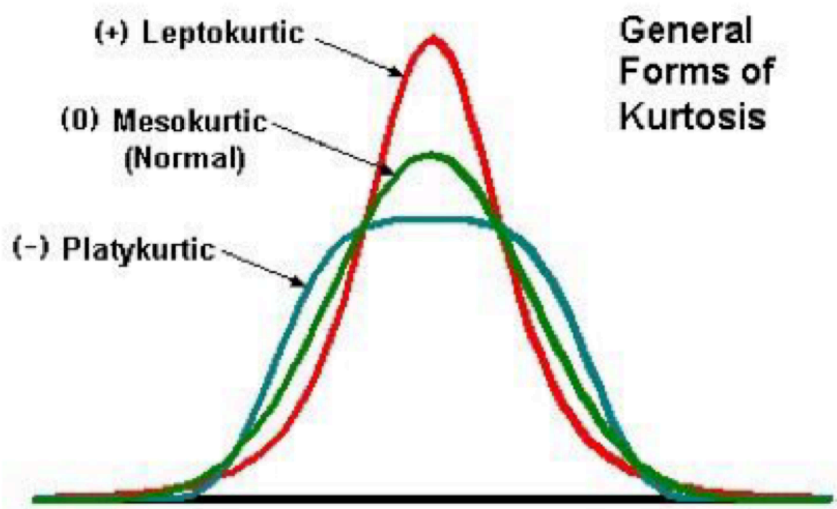


3.2 Kurtosis

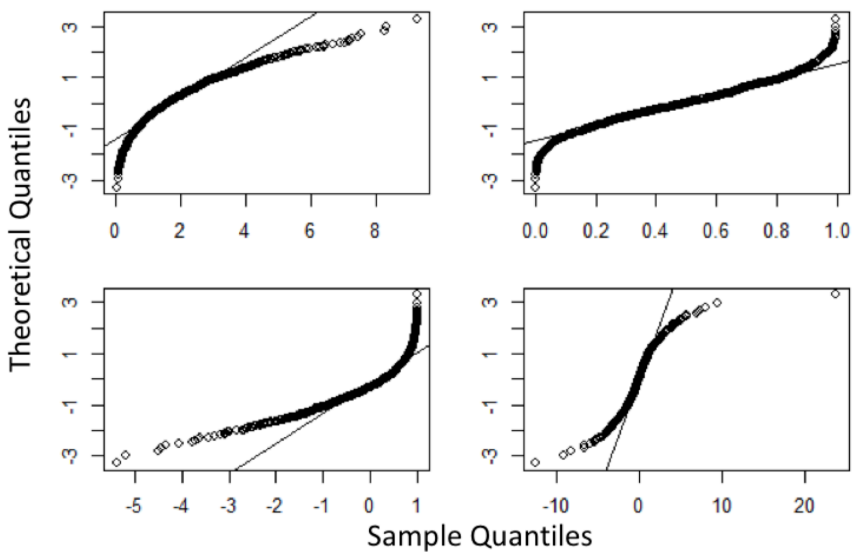
Kurtosis measures how concentrated the data is around the mean, or how heavy the tails are. You can only measure kurtosis (meaningfully) if the distribution has 0 skew.

Kurtosis of a distribution is measured against the kurtosis of the Normal distribution, which is 3.

$$Kur = \mathbb{E}\left[\left(\frac{X-\mu}{\sigma}\right)^3\right] - 3$$



3.3 QQ Plot



- Top left: The sample quantiles spread further on the right than the theoretical quantiles, which implies a **right/pos skewness**
- Top right: The sample quantiles for the tails are very concentrated compared to the theoretical quantiles, which implies the data has short/finite tails implying it is **platykurtic/neg kurtosis**
- Bottom left: The sample quantiles spread further on the left than the theoretical quantiles, which implies a **left/neg skewness**
- Bottom right: The sample quantiles for the tails are very spread compared to the theoretical quantiles, which implies the data has long/heavy tails implying it is **leptokurtic/pos kurtosis**

3.4 Heavy Tail Distributions

A distribution $f(x)$ is said to have:

- Exponential tails (short/finite tails) if

$$f(x) \propto e^{-x/\lambda}$$

- Polynomial tails (long/heavy tails) if

$$f(x) \propto x^{-(1+\alpha)}$$

Where α is the tail index

The smaller the tail index, the heavier the tail

Heavy tailed distributions can also have infinite moments (including the mean!)

$$\mathbb{E}(X^k) = \infty \text{ for } k \geq \alpha$$

So if the tail index is high, the tail is lighter, and the first α moments exist.

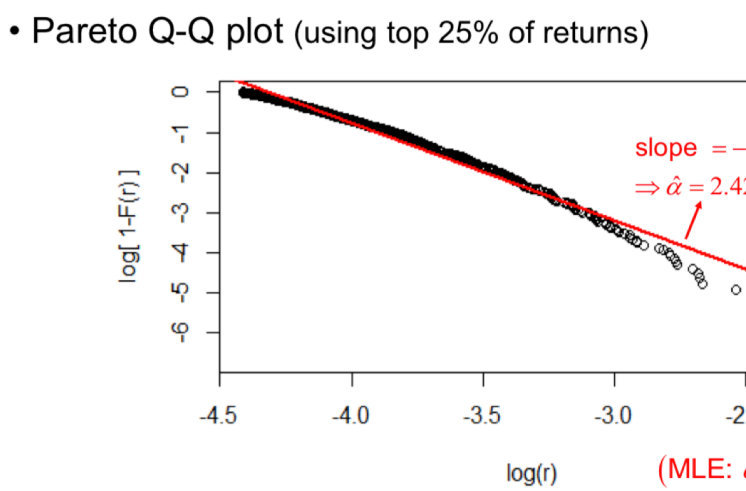
The tail index can be estimated using: - MLE approximation, which takes the derivative log of the product of n the distributions when it equals 0, and solving for α - $\hat{\alpha} = \frac{n}{\sum_{i=1}^{n} \ln(r_i/r_{min})}$ -

$$L(\alpha) = \prod_{i=1}^{n} [f(x)]$$

$$\log(L(\alpha)) = l(\alpha) = \sum_{i=1}^{n} f(x)$$

$$l'(\alpha) = \sum_{i=1}^{n} f'(x) = 0$$

- Pareto Q-Q plot of the empirical CDF and the returns in log x log scale - α is



the estimated slope of the line of best fit

3.4.1 Common Heavy Tail Distributions

3.4.1.1 Pareto

$$f(x) = \frac{c\ x^{-(1+\alpha)}}{l^{\alpha}},\ x \geq l$$

Can be used to model **absolute** returns above a cutoff r_{min}

$$\bar{F}(r) = \left(\frac{r}{r_{min}}\right)^{\alpha} \quad \forall\ r > r_{min}$$

3.4.1.2 Cauchy

$$\begin{aligned} f(x) &= \frac{1}{\pi(1+x^2)} \\ &= \frac{1+x^{-(\alpha+1)}}{\pi} \implies \alpha = 1 \end{aligned}$$

3.4.1.3 Students t

$$f(x) = \frac{\Gamma\left(\frac{v+1}{2}\right)}{\sqrt{v\pi}\Gamma(v/2)}\left(1+\frac{x^2}{v}\right)^{-\frac{v+1}{2}}$$

Where $\alpha = v =$ degrees of freedom

The Students t distributions offers a tractable heavy-tail model of the entire return distribution (not just the tail). It is typically adjusted for the location and scale:

$$Y = \mu + \sigma X \quad \text{where } X \sim t(df = v)$$

3.4.2 Stable Distributions

If we let log returns $r_i \sim$ heavy tail distribution with $0 < \alpha < 2$ (the first and second moment - mean/variance exist)

The aggregate return $r_{1\to n} = r_1 = r_2 + ... r_n \sim$ Stable Distribution

3.4.2.1 Generalized Central Limit Theorem

Stable distributions have no closed form expression, although the share heavy tails and the same tail index α as it's individual distributions.

As you sum independent, **STABLE** random variables, the sum will follow a Stable Distribution.

3.4.2.2 Measuring tail behaviour

We use the complementary CDF

$$\bar{F}(x) = 1 - F(x) = P(X > x) \sim x^{-\alpha}$$

3.5 Mixture Distributions

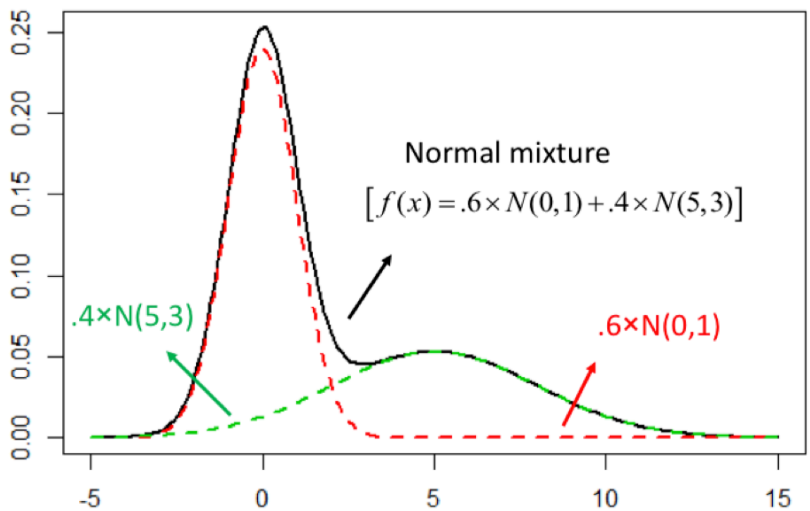
We can generate a random variable using one of out a selection of a family of distributions, choosing the distribution using another distribution.

They are easy to generate, but harder to work with analytically.

We can select from a discrete and finite set of distributions, or of a continuous family of distributions (possibly countable) known as compound distributions.

Example:

$$\text{Generate RV from: } \begin{cases} N(0,1) & p = 60\% \\ N(5,3) & p = 40\% \end{cases}$$



3.5.0.1 Normal scale mixture

$Y = \mu + \sqrt{V} \cdot Z$ Where V is a RV with non-negative mixing distribution and represents a random sd of Y.

Example (probably don't need to memorize…)

$$\text{t-dist } t = Z\sqrt{v/W} \quad \text{where } W \sim \chi^2(df = v)$$

$$\text{GARCH model } r_t = \mu + \sigma_t Z_t \quad \text{where } \sigma_t^2 = \omega + \sum_{i=1}^{p} a_i r_{t-i}^2 + \sum_{j=1}^{q} \beta_j \sigma_{t-j}^2$$

Chapter 4

Problem Set 1

4.1 Q1

4.1.1 Data Prep

Loading libraries required:

```
library(tidyverse)
library(ggplot2)
library(plotly)
library(lubridate)
```

Settings column names for our data (as per the demo file)

```
msg_columns <- c( "Time" , "Type" , "OrderID" ,
                  "Size" , "Price" , "TradeDirection" )
ordr_columns <- c("ASKp1" , "ASKs1" , "BIDp1",  "BIDs1")

# Levels
nlevels = 10;
# naming the columns of data frame
if (nlevels > 1)
{
    for ( i in 2:nlevels )
    {
        ordr_columns <- c (ordr_columns,paste("ASKp",i,sep=""),
                           paste("ASKs",i,sep=""),
                           paste("BIDp",i,sep=""),
                           paste("BIDs",i,sep=""))
```

```
    }
}
```

Reading in the data

```
message = read_csv(paste0(path,"/AMZN_2012-06-21_34200000_57600000_message_10.csv"),
                   col_names=msg_columns)

order_book = read_csv(paste0(path,"/AMZN_2012-06-21_34200000_57600000_orderbook_10.csv"
                      col_names=ordr_columns)
```

```
head(message)
```

```
## # A tibble: 6 x 6
##     Time  Type  OrderID  Size   Price TradeDirection
##    <dbl> <dbl>    <dbl> <dbl>   <dbl>          <dbl>
## 1 34200.     5        0     1 2238200             -1
## 2 34200.     1 11885113    21 2238100              1
## 3 34200.     1  3911376    20 2239600             -1
## 4 34200.     1 11534792   100 2237500              1
## 5 34200.     1  1365373    13 2240000             -1
## 6 34200.     1 11474176     2 2236500              1
```

```
head(order_book)
```

```
## # A tibble: 6 x 40
##     ASKp1 ASKs1  BIDp1 BIDs1  ASKp2 ASKs2  BIDp2 BIDs2  ASKp3 ASKs3  BIDp3 BIDs3
##     <dbl> <dbl>  <dbl> <dbl>  <dbl> <dbl>  <dbl> <dbl>  <dbl> <dbl>  <dbl> <dbl>
## 1 2239500   100 2.23e6   100 2.24e6   100 2.23e6   200 2.24e6   220 2.23e6   100
## 2 2239500   100 2.24e6    21 2.24e6   100 2.23e6   100 2.24e6   220 2.23e6   200
## 3 2239500   100 2.24e6    21 2.24e6    20 2.23e6   100 2.24e6   100 2.23e6   200
## 4 2239500   100 2.24e6    21 2.24e6    20 2.24e6   100 2.24e6   100 2.23e6   100
## 5 2239500   100 2.24e6    21 2.24e6    20 2.24e6   100 2.24e6   100 2.23e6   100
## 6 2239500   100 2.24e6    21 2.24e6    20 2.24e6   100 2.24e6   100 2.24e6     2
## # ... with 28 more variables: ASKp4 <dbl>, ASKs4 <dbl>, BIDp4 <dbl>,
## #   BIDs4 <dbl>, ASKp5 <dbl>, ASKs5 <dbl>, BIDp5 <dbl>, BIDs5 <dbl>,
## #   ASKp6 <dbl>, ASKs6 <dbl>, BIDp6 <dbl>, BIDs6 <dbl>, ASKp7 <dbl>,
## #   ASKs7 <dbl>, BIDp7 <dbl>, BIDs7 <dbl>, ASKp8 <dbl>, ASKs8 <dbl>,
## #   BIDp8 <dbl>, BIDs8 <dbl>, ASKp9 <dbl>, ASKs9 <dbl>, BIDp9 <dbl>,
## #   BIDs9 <dbl>, ASKp10 <dbl>, ASKs10 <dbl>, BIDp10 <dbl>, BIDs10 <dbl>
```

Setting up the X-axis (time of day) and Y-axis (price) variables

```r
# Trading hours (start & end)
startTrad = 9.5*60*60        # 9:30:00.000 in ms after midnight
endTrad = 16*60*60         # 16:00:00.000 in ms after midnight

# Define interval length
freq = 5*60;    # Interval length in ms 5 minutes

# Number of intervals from 9:30 to 4:00
no_int= (endTrad-startTrad)/freq
```

```r
# Convert prices into dollars
#    Note: LOBSTER stores prices in dollar price times 10000

# Sample rows closest to 5 minute increments
message_data = message %>%
  mutate(Price = Price / 10000,
         idx = row_number(),
         five_min_grp = ((Time - startTrad) %/% freq) * freq + startTrad) %>%
  group_by(five_min_grp) %>%
  filter(row_number() == 1) %>%
  ungroup() %>%
  arrange(idx) %>%
  # Format the Time column to be understandable,
  # needed to add 4 hours to be in right time zone
  mutate(X_Time = (Time + 4*60*60) %>%
           as.POSIXct.numeric(origin = '2012-06-21') %>%
           lubridate::round_date(unit='minute'))

order_data = order_book %>% mutate(idx = row_number())
price = order_data %>%
  select(contains("p"),idx) %>%
  pivot_longer(!idx,
               # First column is Order Type, takes first capture group, second
               # capture group is excluded, third capture group of levels
               # becomes col names, indicated by .value
               names_to = c("Order Type", ".value"),
               names_pattern = "(...)(?:p)(\\d{1,2})") %>%
              # need to pivot longer again, but excluding Order Type and row idx
              # this pivots the levels into its own column
  pivot_longer(!c(`Order Type`,idx),
               names_to = c("Level"),
               values_to= "Price")

quantity = order_data %>%
  select(!contains("p"), idx) %>%
```

```
  pivot_longer(!idx,
               names_to = c("Order Type", ".value"),
               names_pattern = "(...)(?:s)(\\d{1,2})") %>%
  pivot_longer(!c(`Order Type`,idx),
               names_to = c("Level"),
               values_to= "Quantity") %>%
  select(Quantity)

order_longer = price %>% bind_cols(quantity) %>%
  mutate(Limit_Price = Price / 10000) %>% filter(idx %in% message_data$idx) %>%
  select(-Price)

plt_data =  order_longer %>%
  left_join(message_data %>% select(Time, Price, idx, X_Time), by="idx")
```

4.1.2 Plot

```
(plt_data %>%
  ggplot(aes(x=X_Time)) +
  geom_point(aes(y=Limit_Price, size=Quantity, color=`Order Type`)) +
  geom_line(aes(y=Price)) +
  scale_color_manual(values=c("red2", "green3")) +
  scale_x_datetime(breaks = seq(min(plt_data$X_Time), max(plt_data$X_Time),
                                by = 30 * 60), date_labels="%H:%M") +
  scale_y_continuous(labels = scales::dollar_format(prefix="$")) +
  labs(title = "Evolution of AMZN Limit Order Book at 5-minute intervals",
       y="Price($)",
       x="Time of Day")) %>%
  ggplotly()
```

4.2 Q2

R (Daily Return Data) Download the closing and adjusted closing daily prices for Apple Inc. (symbol APPL) for the previous year (Jan 1, 2022 to Dec 31, 2022). Use the the function get.hist.quote() from the tseries package.

Loading Data

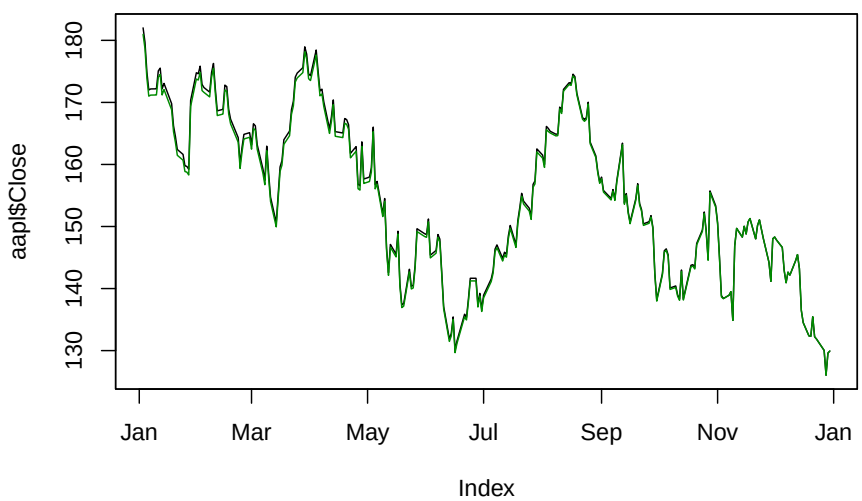
```
library(tseries)
aapl = get.hist.quote('aapl', start="2022-01-01", end="2022-12-31",
                      quote=c("Close", "Adjusted"), compression="d")
```

```
## time series starts 2022-01-03
## time series ends   2022-12-30
```

4.2.1 2.a

Plot the closing and adjusted-closing price series on the same plot. Do you see any differences?

```
plot(aapl$Close)
lines(aapl$Adjusted, col="green4")
```

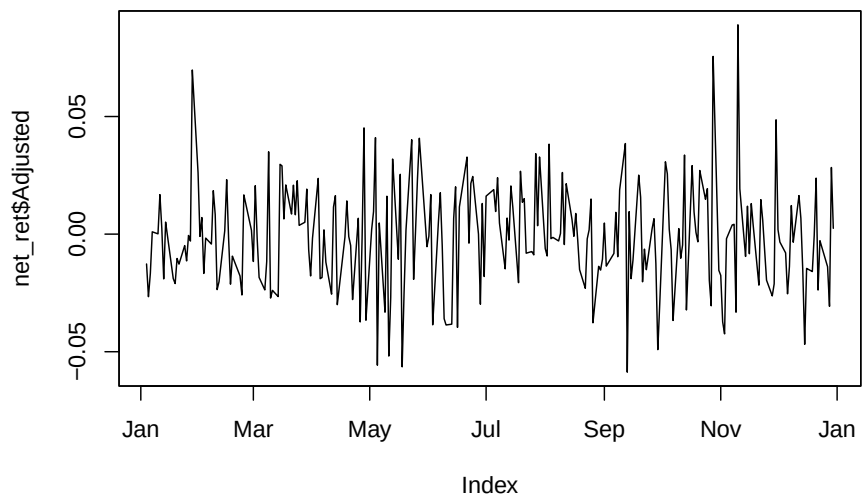


It seems there are differences in the first half of the year.

4.2.2 2.b

Calculate and plot the net returns based on the adjusted closing price.

```
# we shouldn't have a value for Jan 3rd, because that was the first trading day
# no way to calculate return from the day before. time series starts jan 4th
net_ret = aapl / stats::lag(aapl, -1) - 1
log = log(aapl) - log(stats::lag(aapl, -1))
returns = cbind(net_ret, log)
plot(net_ret$Adjusted)
```



4.2.3 2.c

Apple paid a dividend of $0.23 on Nov 4, 2022 (such events are documented here). Calculate the net return of Nov 4, using both the adjusted closing prices (designated way), and the actual closing prices, where you manually adjust for the dividend (i.e., you add the dividend amount to the Nov 4 close price before calculating the log-return). Are the two returns equal?

```
adj_nov4_ret = net_ret["2022-11-04"]$Adjusted[[1]]

act_nov4_ret = (aapl["2022-11-04"]$Close + 0.23)[[1]] /
  aapl["2022-11-03"]$Close[[1]] - 1

log_nov4_ret = log(aapl["2022-11-04"]$Close + 0.23)[[1]] -
  log(aapl["2022-11-03"]$Close)[[1]]

# Net Return using adjusted close
adj_nov4_ret
```

```
## [1] -0.001947378
```

```
# Net Return using actual close
act_nov4_ret
```

```
## [1] -0.001944124
```

```
# Log Return using actual close
log_nov4_ret
```

```
## [1] -0.001946017
```

They are the same up to 5 significant digits.

4.2.4 2.d

Consider a simple momentum strategy whereby: - you buy APPLE stock for 1 day, if its price increased by more than 2% on the previous day, and - you (short) sell APPLE stock for 1 day, if its price decreased by more than 2% on the previous day Calculate and plot the daily and cumulative net returns of this strategy.

```
# If return at t-1 > 2%, buy APPLE for 1 day, sell next day
# If return at t-1 < 2%, short sell APPLE for 1 day, buy back next day

# Dates where AAPL went up 2%
days_plus2 = time(returns[returns$Adjusted.log > 0.02])

# Dates where AAPL went down 2%
days_minus2 = time(returns[returns$Adjusted.log < 0.02])

## records the return of the day after a >2% return
return_2d_pos = stats::lag(returns,1)[days_plus2]
return_2d_neg = stats::lag(returns,1)[days_minus2]
return_2d = rbind(return_2d_pos, return_2d_neg)

net_cum_2d_ret = cbind("Daily Returns" = return_2d$Adjusted.log,
                       "Cumulative Returns" = cumsum(return_2d$Adjusted.log))
(net_cum_2d_ret %>%
  ggplot(aes(x=time(net_cum_2d_ret))) +
  geom_line(aes(y=`Daily Returns`, color="one")) +
  geom_line(aes(y=`Cumulative Returns`,color="two")) +
  scale_color_discrete(name = "Return Type",
                       labels = c("Daily Net Return",
                                  "Cumulative Net Return")) +
```

```
  labs(title="Momentum strategy returns")) %>%
  ggplotly()
```

4.2.5 2.e

Find the maximum drawdown of the strategy, i.e. the biggest drop in cumulative net returns from their highest point. You can use the maxdrawdown() function from the tseries package.

```
maxdrawdown(net_cum_2d_ret$`Cumulative Returns`)
```

```
## $maxdrawdown
## [1] 0.3460492
##
## $from
## [1] 58
##
## $to
## [1] 247
```

4.3 Q3

A stock has an expected annual log-return of 8% and an annual volatility (i.e. st. deviation of annual log-return) of 20%. You want to model the daily log-return using a Normal Random Walk model. What should the mean and variance of the Normal distribution of the daily log-returns be, so that your model matches the annual parameters above? Assume that a calendar year has 252 trading days of daily returns.

```
# num trading days
num_days = 252
```

4.3.1 Mean of daily log-return

Question implies $log(\frac{P_{252}}{P_0}) = 0.08$ We want the daily mean, so $log(\frac{P_{252}}{P_{251}}) + ... + log(\frac{P_1}{P_0}) = 0.08$ assuming $log(\frac{P_{252}}{P_{251}}) = ... = log(P_1/P_0)$ implies $log(P_t/P_{t-1}) = 0.08/252 = 0.0003174603$

Can also use formula from class

$$E[r_{1-n}] = n\mu\mu = E[r_{1-n}]/n$$

```
daily_mean = 0.08 / 252
```

4.3.2 Volatility of daily log-return

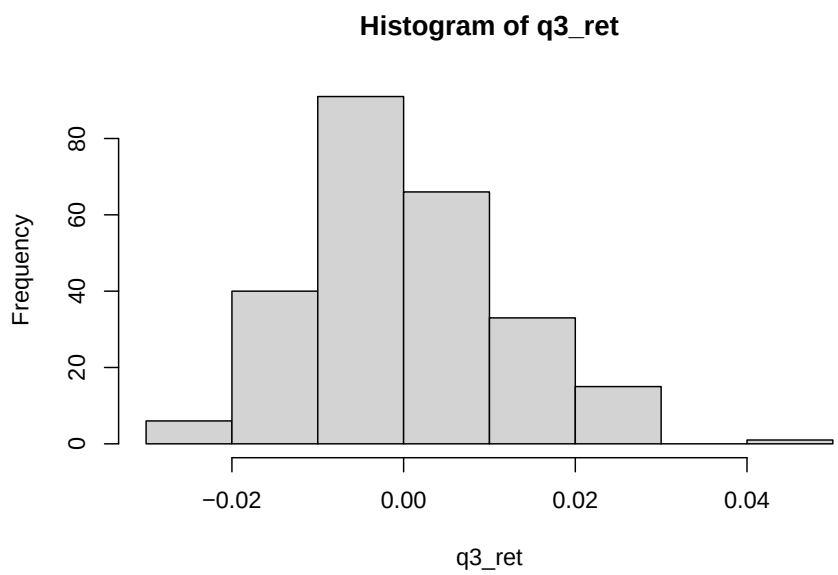
Using formula from class

$$\sqrt{Var[r_{1-n}]} = \sqrt{n}\sigma\sigma = \frac{sd[r_{1-n}]}{\sqrt{n}}$$

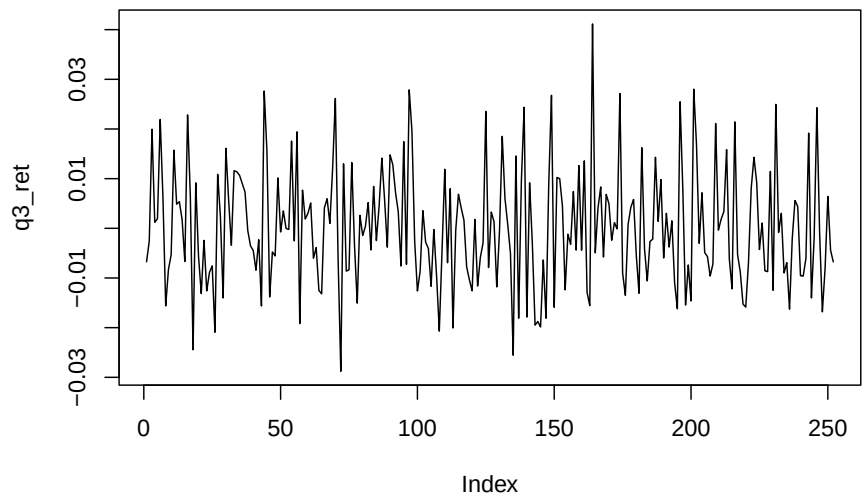
```
# volatility/variance
std_dev = 0.20
daily_std = std_dev / sqrt(num_days)
```

4.3.3 Plot

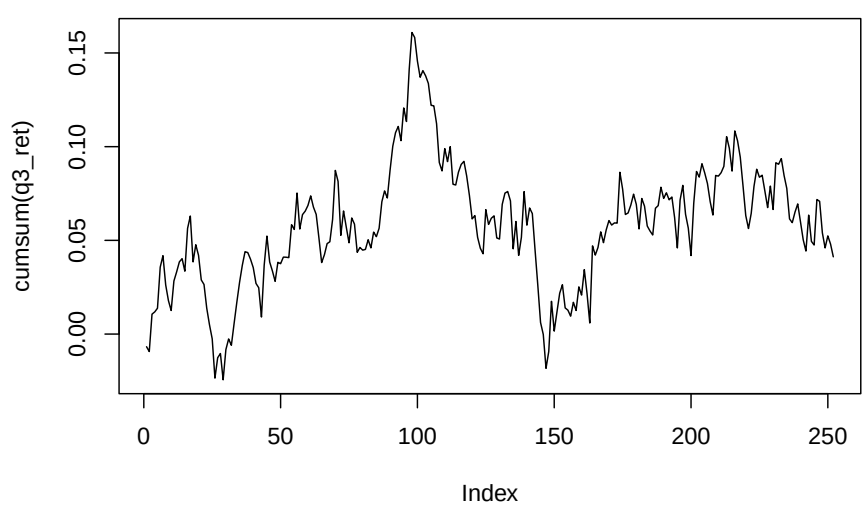
```
set.seed(123)
q3_ret = rnorm(num_days, daily_mean, daily_std)
hist(q3_ret)
```



```
plot(q3_ret, type="l")
```




```
plot(cumsum(q3_ret), type="l")
```



4.4 Q4

Assume that the daily log returns on a stock with current price $S_0 = 100$ are i.i.d. with a Normal distribution with mean `0.0003` and standard deviation `0.01`.

4.4.1 4.a

Calculate the probability that the price will be above $120 after 50 days.

```
mean = 0.0003
std = 0.01
s_0 = 100
days = 50

# using the normal distribution, just need to find params for the 50 log-returns
# and then find prob above 120 (ie. return is 20%)
q4_50_day_mean = days * mean
q4_50_day_std = std * sqrt(days)

# probability
prob_4a = pnorm(.20, q4_50_day_mean, q4_50_day_std, lower.tail = FALSE)
prob_4a
```

```
## [1] 0.004444485
```

4.4.2 4.b

Calculate the probability that the price will increase (i.e., have +ve return) in 30 or more of the next 50 days. Note: You will have to use (pnorm(), pbinom()) to find the numerical values of these probabilities.

```
prob_4b = pbinom(30, 50, pnorm(0, mean, std, lower.tail = FALSE))
prob_4b
```

```
## [1] 0.9177322
```

Chapter 5

Problem Set 2

5.1 Q1

An MLE can be derived from finding the derivative of the log $L(\alpha)$, $l(\alpha)$ function, where

$$L(\alpha) = \prod_{i=1}^{n} \left[\frac{\alpha x_i^{-(\alpha+1)}}{\ell^{-\alpha}} \right]$$

$$\log L(\alpha) = l(\alpha) = \sum_{i=1}^{n} \left[\log(\alpha) - (\alpha+1)\log(x_i) + \alpha \log(\ell) \right]$$

$$l'(\alpha) = \sum_{i=1}^{n} \left(\frac{1}{\alpha} - [\log(x_i) + \log(\ell)] \right) = 0$$

$$0 = \frac{n}{\alpha} - \sum_{i=1}^{n} (\log(x_i) - \log(\ell))$$

$$\frac{n}{\alpha} = \sum_{i=1}^{n} (\log(x_i) - \log(\ell))$$

$$\frac{1}{\alpha} = \frac{1}{n} \left[\sum_{i=1}^{n} \log\left(\frac{x_i}{\ell}\right) \right]$$

$$\therefore \hat{\alpha} = \left(\frac{1}{n} \left[\sum_{i=1}^{n} \log\left(\frac{x_i}{\ell}\right) \right] \right)^{-1}$$

5.2 Q2

5.2.1 Part A

Heavy tail distributions are those with polynomial tails which follow by

$$f(x) \propto x^{-(1+\alpha)}$$

where smaller $\alpha \to$ heavier tails

We say that distribution F is heavy tailed if the integral on the positive half of $\mathbb{R}^+$ is infinite for some t

$$m_X(t) = E(e^{tX}) = \left[1 + tE(X) + \frac{(tE(X))^2}{2!} + \frac{(tE(X))^3}{3!} + ... + (tE(X))^n/n! + ...\right]$$

$\widehat{}$ not complete

5.2.2 Part B

$$\begin{aligned}
\phi_X(t) &= \mathbb{E}[e^{itX}] = \mathbb{E}[\cos(tX) + i\sin(tX)] \\
&= \mathbb{E}[\cos(tX)] + i\mathbb{E}[\sin(tX)] \\
\mathbb{E}[\sin(tX)] &= \int_{-\infty}^{\infty} \sin(tX) f(x)\, dx \\
&= \int_{-\infty}^{0} \sin(tX) f(x)\, dx + \int_{0}^{\infty} \sin(tX) f(x)\, dx \\
&= -\int_{0}^{\infty} \sin(tX) f(x)\, dx + \int_{0}^{\infty} \sin(tX) f(x)\, dx \\
&= 0 \\
\therefore \phi_X(t) &= \mathbb{E}[\cos(tX)] \text{ which only takes real values}
\end{aligned}$$

5.2.3 Part C

$$\begin{aligned}
\phi_X(t) &= \mathbb{E}[e^{itX}] \\
\phi_Y(t) &= \mathbb{E}[e^{itY}] \\
\phi_Z(t) &= \mathbb{E}[e^{itZ}] \\
&= \mathbb{E}[e^{it(X+Y)}] \\
&= \mathbb{E}[e^{itX} e^{itY}] \\
&= \mathbb{E}[e^{itX}]\mathbb{E}[e^{itY}] \quad \text{As they are independent} \\
&= e^{-c|t|^\alpha} \cdot e^{-d|t|^\alpha} \\
&= e^{-(c+d)|t|^\alpha}
\end{aligned}$$

Which shows that $\phi_Z(t)$ shares the same form as X and Y.

5.2.4 Part D

$$
\begin{aligned}
\phi_X(t) &= \mathbb{E}[e^{itX}] = e^{-|t|} \\
\phi_{\bar{X}}(t) &= \mathbb{E}[e^{it\bar{X}}] \\
&= \mathbb{E}\left[e^{it\frac{1}{n}(\sum_{i=1}^{n} X_i)}\right] \\
&= \mathbb{E}\left[e^{it\frac{1}{n}X_1} e^{it\frac{1}{n}X_2} \dots e^{it\frac{1}{n}X_n}\right] \\
&= \mathbb{E}\left[e^{it\frac{1}{n}X_1}\right] \mathbb{E}\left[e^{it\frac{1}{n}X_2}\right] \dots \mathbb{E}\left[e^{it\frac{1}{n}X_n}\right] \\
&= [e^{-|t|/n}]^n \\
&= e^{-|t|/n \cdot n} = e^{-|t|} \quad \text{Which is exactly the } t(1) \text{ distribution}
\end{aligned}
$$

5.2.5 Part E

According to theory, $\mathbb{E}[X^k] = \infty$ for any $k \geq \alpha$ where α is the tail index.

If the tail index is greater than 2, then the second order moment exists, implying both the mean and variance both exist/are finite.

By definition, a sum of stable distributions would converge to a stable distribution as shown by the characteristic functions above, but according to the Central Limit Theorem, any sum of iid random variables with finite mean and variance will produce a Normal distribution.

5.3 Q3

later